

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch : IT

Semester : VII

Subject Code & Name:(BTITPE705D)Distributed Computing

Max Marks: 60

Date:09-02-2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) What is transparency in distributed systems? What are the different forms of transparency?	Remember	06
B) What is distributed system? Explain the design issues of distributed system in detail.	Remember	06
C) What is a network? What are the types of switching techniques?	Remember	06
Q.2 Solve Any Two of the following.		12
A) What is virtualization in distributed computing?	Remember	06
B) What is process migration? What are two types of process migration? What are the steps involved in process migration?	Remember	06
C) Explain the working of Remote Procedure Call (RPC) in distributed computing. Also explain the types of RPC.	Understand	06
Q.3 Solve Any Two of the following.		12
A) What is Domain Name System (DNS)? Explain the working of domain name system in detail.	Remember	06
B) Explain X.500 directory service in detail.	Understand	06
C) What is an election algorithm? Explain Bully algorithm in detail.	Remember	06
Q.4 Solve Any Two of the following.		12
A) Explain Christians' algorithm for distributed clock synchronization.	Understand	06
B) What are distributed commit protocols? Explain types of distributed commit protocols.	Remember	06
C) What is a Client-Centric Model? Explain the following:	Remember	06
1. Monotonic Reads		2. Monotonic Writes

3. Read Your Writes
5. Eventual Consistency

4. Writes Follow Reads

- Q. 5 Solve Any Two of the following.** **12**
- A) What is Peer-to-Peer (P2P) network? What are the types of P2P networks? Remember **06**
How does a P2P network work?
- B) Explain Pastry overlay network in detail. Understand **06**
- C) Differentiate between RMI and CORBA. Analyze **06**

***** End *****